

Assignment 2:

Subject: MPCA

Chapter 2: 8085 MP

- 1. Explain, in brief, 8085 Microprocessor with its block diagram.**
- 2. Explain the 8085 pin configuration in brief.**
- 3. What are accumulator and PC? Explain in short.**
- 4. Define instruction. Classify 8085 instruction set and discuss in brief.**
- 5. Explain the address modes in 8085.**
- 6. What are instruction cycle, machine cycle and T-state and Timing diagram?**
- 7. Explain different types of machine cycle.**
- 8. Explain opcode fetch machine cycle with its timing diagram.**
- 9. Explain memory read/write machine cycle with its timing diagram.**
- 10. Explain IO read/write machine cycle with its timing diagram.**
- 11. Draw the timing diagram for MOV A,B**
- 12. Draw the timing diagram for MVI A, 43H**
- 13. Draw the timing diagram for STA 2050H**
- 14. Write an ALP for subtracting two 8-bit numbers stored in 2501H and 2502H. The result is to be stored in 2503H.**
- 15. Write an assembly language program to add two 8-bit numbers (F1 and 15), stored at C050H and C051H, with carry.**
- 16. Write a program in Assembly language to interchange(Swap) the contents of two memory location 2100H and 2101H.**
- 17. Define RTL. Consider a system with four 1-bit flipflops. Given RTL code where j, o, h and n are conditions which are mutually exclusive. Draw its hardware implementation.**

$j: M \leftarrow A$
 $o: A \leftarrow Y$
 $h: R \leftarrow M$
 $n: Y \leftarrow R, M \leftarrow R$

18. What are the uses of 8085 MP? Mention the issues in 8085.

Chapter 3: 8086 MP

19. Explain the 8086 MP architecture with its block diagram.
20. Explain the pin configuration of 8086 in brief.
21. Explain different flags used in 8086 MP.
22. Explain different General-purpose registers in 8086 MP.
23. Explain different segment registers in 8086 MP.
24. Explain the addressing modes of 8086 MP.
25. Compare 8085 and 8086 MPs.
26. What is editor, assembler and linker?

Chapter 7: IO organization

27. Differentiate between Serial and Parallel Communication.
28. Differentiate between Serial Interface and Parallel Interface.
29. What is IO operation? Explain in brief about the different modes of IO transfer.
30. Explain Programmed IO with a diagram.
31. Explain Interrupt-driven IO.
32. Explain DMA technique.
33. What are the different types of interrupts? Explain in short.
34. How interrupt processing is done? Explain.

- 35. Explain priority arbitration with block diagram.
- 36. Explain daisy chain arbitration with block diagram.
- 37. What are IO processors? Explain.
- 38. Describe UART.

Chapter 9: Advanced Architecture.

- 39. Differentiate between RISC and CISC.
- 40. Explain RISC instruction pipelining with different stages.
- 41. What is register window? In a RISC processor system, there are 10 global registers and 8 register windows, each having 4 common input registers, 10 local registers, and 4 common output registers. How many registers are in this CPU? Also show the diagrammatic representation.
- 42. Explain Flynn's classification for multiprocessors.
- 43. Explain different topologies for connection of multiprocessors.
- 44. Explain UMA and NUMA.
- 45. How can memory be accessed or shared in multiprocessors architecture? Explain each of them.
- 46. What is cache coherence? Explain.
- 47. Introduce multicore architecture with necessary diagrams.